May 28, 2026

ENGR. EMMANUEL L. FERRER, PH.D.
Assistant Director for Academic Affairs
Technological University of the Philippines – Taguig

Thru:

ENGR. RAYMUNDO M. LOZADA, PECE
Dean, College of Engineering and Technology

DR. RICO S. SANTOS
Department Head, College of Engineering and Technology

To the Office of Academic Affairs,

SUBJECT: Proposal: Haki — Offline Student Information Assistant for TUP Taguig

OFFLINE-FIRST **SOURCE-REFERENCED** **ADMIN-GOVERNED**

WHERE THIS IDEA CAME FROM

I began developing Haki in early May 2026 as a personal portfolio project while preparing to apply for AI/ML engineering roles in the coming months. The goal was straightforward: build something genuinely useful for students here at TUPT, while demonstrating real-world skills to future employers. To put it simply — a first-year student who does not know where the Registrar's office should be able to find it, as well as its current procedures in seconds, without an internet connection.

The idea to formally offer Haki to the University was encouraged when Ma'am Mara raised the possibility of posting my participation in the **MIT Sloan AI Club Global Hackathon** — an international AI competition hosted by MIT — on the MTICS group page as a marketing strategy for IT at TUPT. That conversation sparked a broader thought in my mind: if we are already talking about marketing, why not go further?

To my knowledge, among Philippine state universities and private colleges, none currently offers an app where students can ask campus-related questions, receive AI-assisted guidance, and access relevant information fully offline. If TUPT adopts Haki, **TUP Taguig has the opportunity to pioneer this in the AI age** — a genuine milestone that, with the right promotion and advertisement, positions the University as a leader in responsible, student-led innovation.

WHAT HAKI DOES

Haki is an offline-first application built specifically for TUP Taguig students. In its current working state, it can:

- Provide access to locally saved, administration-approved campus information, with every response citing its source.
- Display an in-app TUPT campus map — showing buildings, room numbers, office locations, faculty rooms, and other key areas within the campus.
- Support manual search functionality within the app when AI responses are unavailable or insufficient.
- Support optional on-device AI chat on compatible devices using a local model

Screenshots of the core features are included in Annex A for your visual review.

The current development builds use a mix of **publicly available campus information** — such as materials posted at offices in the administration building — and **placeholder content** for development and testing purposes. This is not final and should not be treated as official information for deployment. I recommend that the Administration review and designate the appropriate content before any public release.

I also recommend that the **official pages of the different University offices** remain the primary source for real-time updates and announcements. Haki is intended to serve as a convenient offline reference — not a replacement for official communication channels.

WHAT I AM OFFERING

I am offering Haki to TUPT **at no development charge**. This is a personal project, and my goal is simply to see it benefit the people I built it for.

The Administration holds full authority over what information is made available through the app. I defer entirely to the Administration on which offices, procedures, policies, and contact details may be included, and how their accuracy is maintained and updated over time.

IMPLEMENTATION AND STEWARDSHIP

If adopted, for long-term continuity, I recommend the following:

- **Assign student/staff maintainers** — a responsible TUPT office designates students, under appropriate supervision, to handle future content updates and technical continuity.
- **Guided turnover** — I will personally guide the first batch of maintainers, so they understand the application structure and the importance of using only administration-approved information.

Should the University adopt Haki, I will remain available for any assistance or clarifications I am able to provide. A proper handover with documentation will be completed to ensure continuity does not depend solely on my availability after graduation.

Haki is currently available on **Android only**. An iOS release is possible.

APP STORE DISTRIBUTION AND EXPENSES

Publishing Haki through official app stores would make it reliably accessible to all students. The associated costs are:

- **Google Play Store:** US$25 one-time developer registration fee
- **Apple App Store:** US$99 annual Apple Developer Program membership fee (eligible educational institutions may qualify for a fee waiver)

My recommendation is to set a **PHP 49 one-time download fee** on the stores. This serves two purposes: it helps cover the Apple App Store's annual fee, and any remaining net proceeds — after platform deductions — can serve as additional funds for the **University Student Government (USG)** for student events and initiatives.

However, if the University is able to cover the annual fee independently, the app could be made available to students at **no cost**. Either way, **this decision should come from the Administration in consultation with the USG**, with full transparency in financial governance and formally approved accounting procedures.

I recommend first checking whether an institutional publisher account or educational fee waiver is applicable before committing to any expense.

DATA PRIVACY

Haki is designed as an offline-first application and does not collect, store, or transmit personal data to external servers. Any institutional use of the application shall remain subject to the **Data Privacy Act of the Philippines (Republic Act No. 10173)** and all applicable University policies on data governance and information security. All data live on the user's device only.

THE ADMINISTRATION'S ROLE

This is a proposal, not a requirement. All decisions — whether to adopt Haki, what content it carries, how it is distributed, and how it is governed — rest entirely with the Administration. My role is to build, hand over, and support.

The only benefit I respectfully request, if Haki is implemented or promoted by TUPT, is the right to **claim credit for originally developing the application** — to reference it in my portfolio, resume or CV, and during job interviews, without disclosing any restricted University information.

REQUESTED ADMINISTRATIVE ACTION

1. Consider whether institutional adoption of Haki is appropriate for TUP Taguig.

   If will be adopted,

2. Identify the official campus information that may be stored and made available through the application, replacing current placeholder content.
3. Advise on the assignment of a responsible office and student maintainers for content updates and long-term app continuity.
4. Advise on whether Google Play or App Store publication should be authorized, and determine the appropriate account, download pricing, and financial governance in consultation with the USG.
5. Confirm an acceptable attribution arrangement for the original developer if Haki is deployed or promoted.

Thank you for considering this proposal. I am ready to walk you through a live demonstration of Haki at your convenience. Please feel free to reach me directly at **joshuaretrita@gmail.com** or **09206427509**, or through Sir Rico or Ma'am Mara. I am happy to cooperate with any evaluation or governance process the Administration may require.

Respectfully submitted,

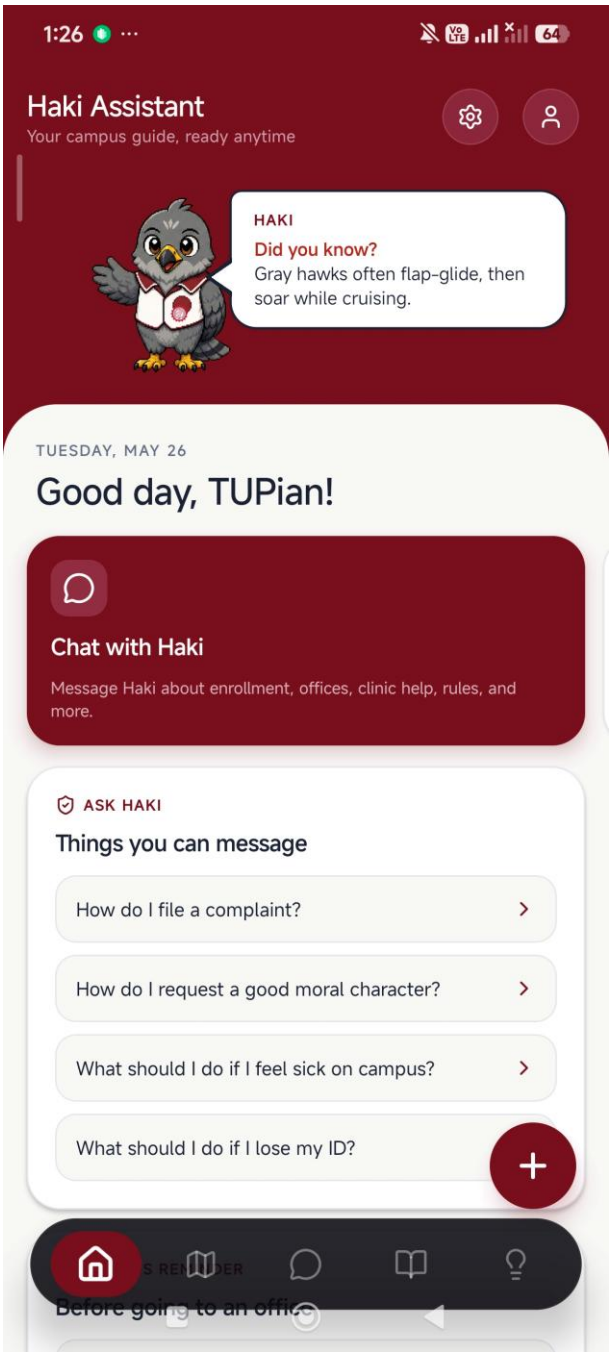
Joshua Retrita
BSIT Student, Academic Year 2025–2026
TUPT-21-0096

Sources: *App store fee statements reviewed on May 24, 2026 — Google Play Console Help (support.google.com/googleplay/android-developer/answer/6112435) and Apple Developer Support (developer.apple.com/support/compare-memberships/).*

ANNEX A — CORE FEATURES OF HAKI

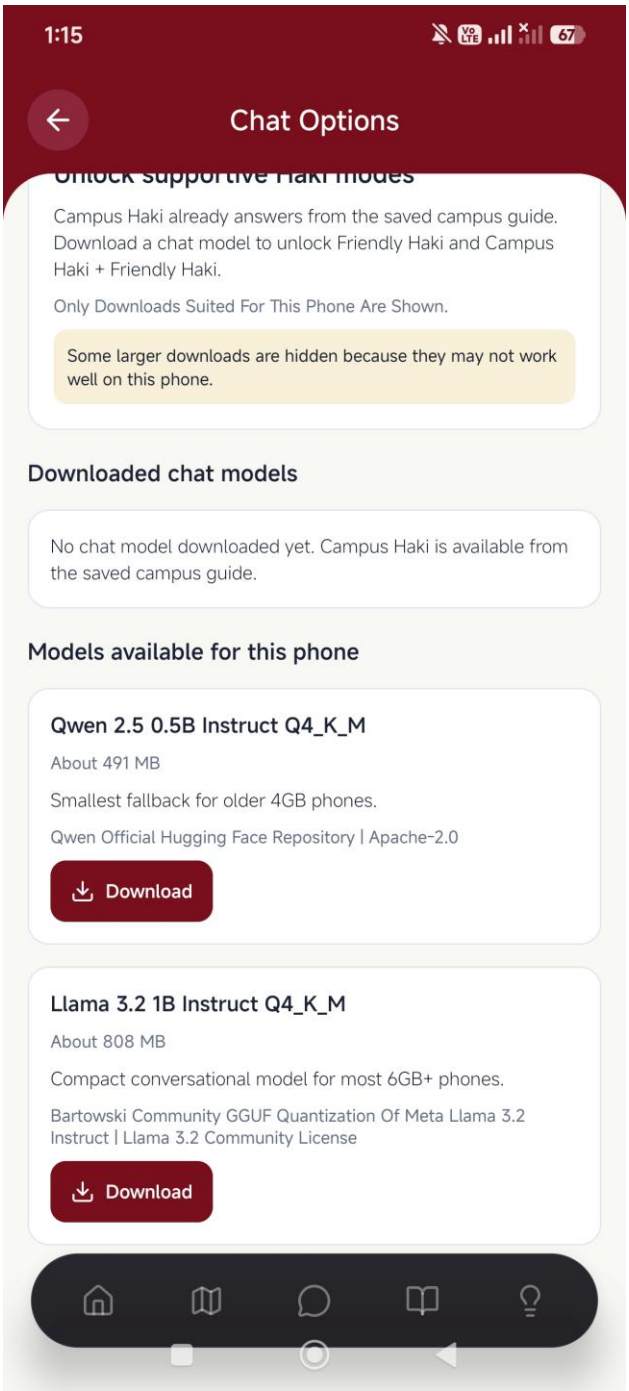
Illustrative development-build interface. Official content and deployment remain subject to TUPT Administration approval. Screenshots should be viewed from left to right.

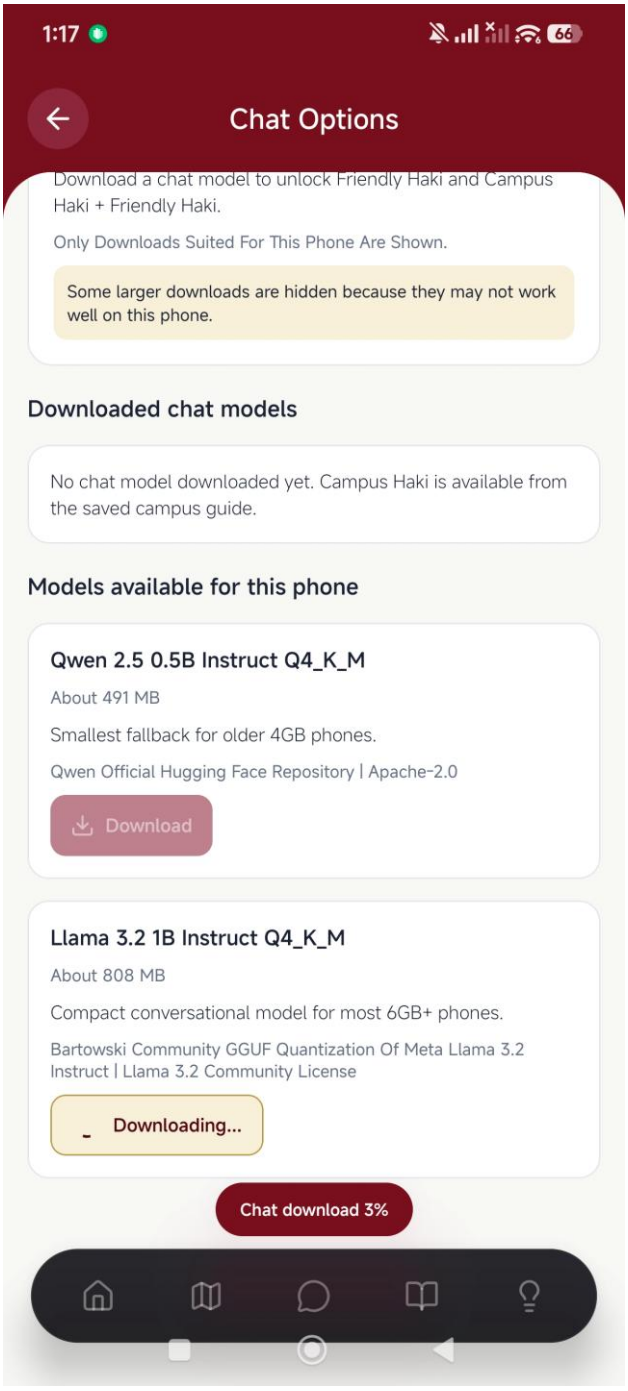
A1 – HOME SCREEN / LANDING PAGE

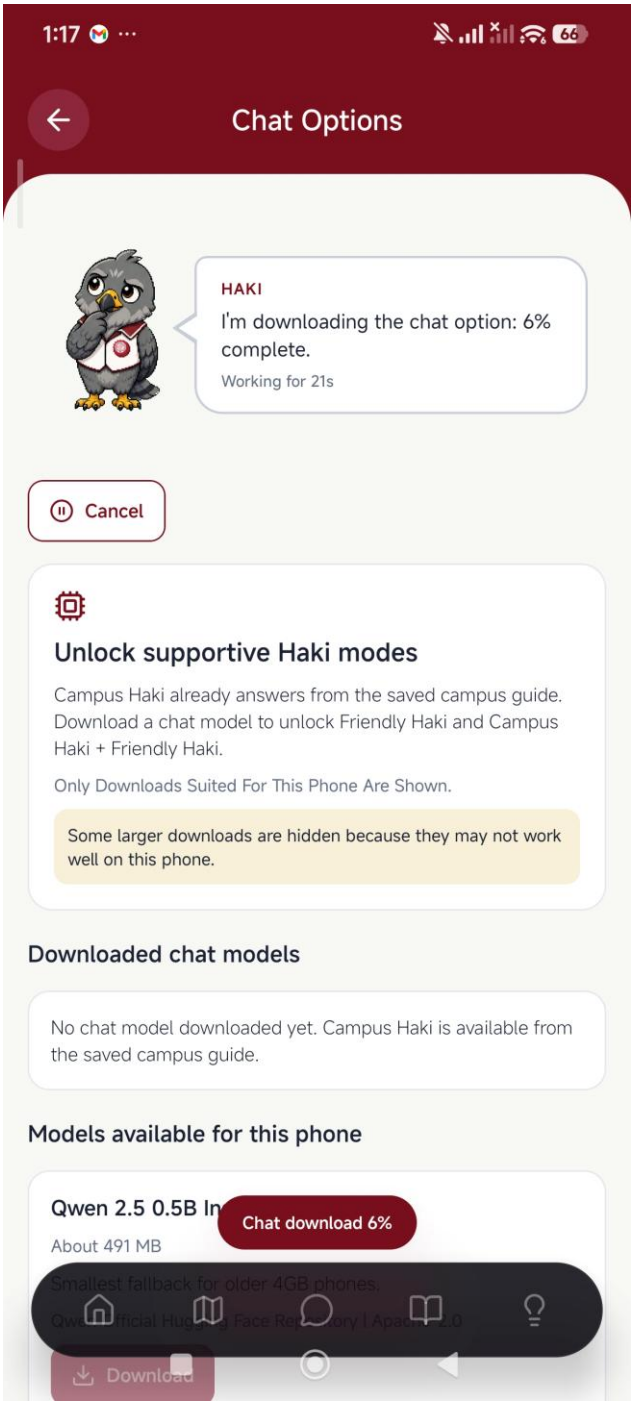


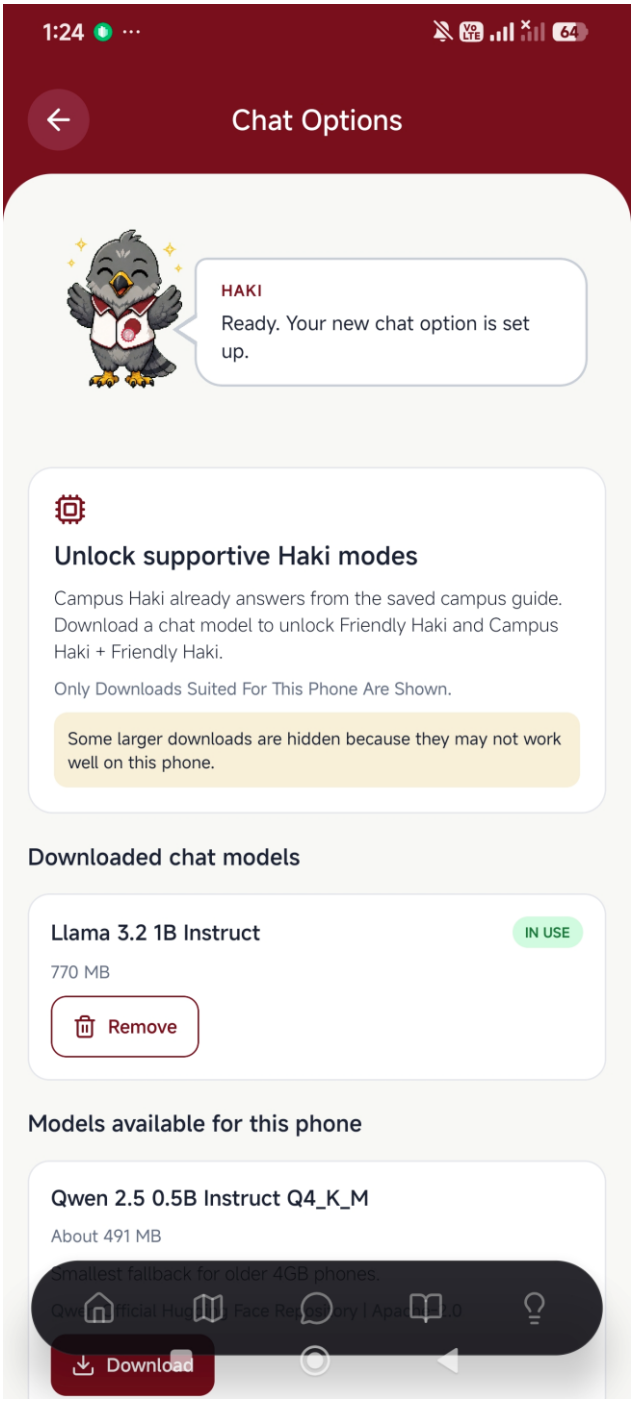
A2 – MODEL SELECTION

Haki will automatically detect capable models based on user's device



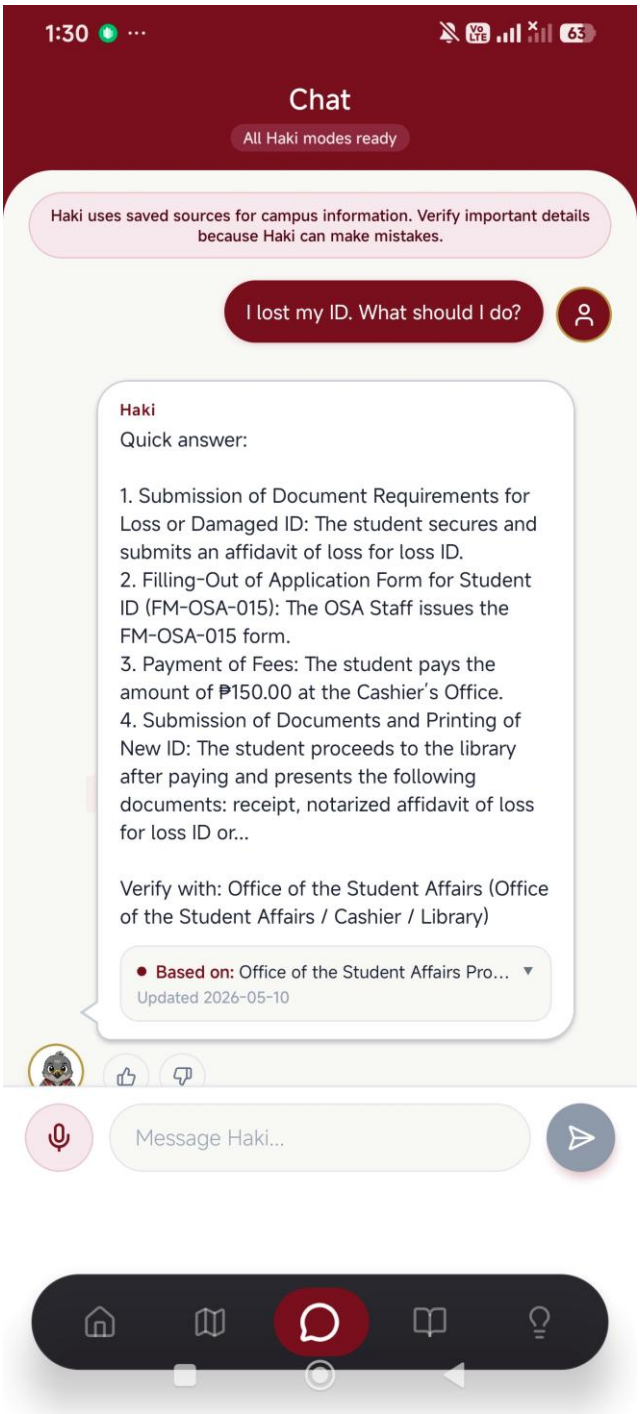


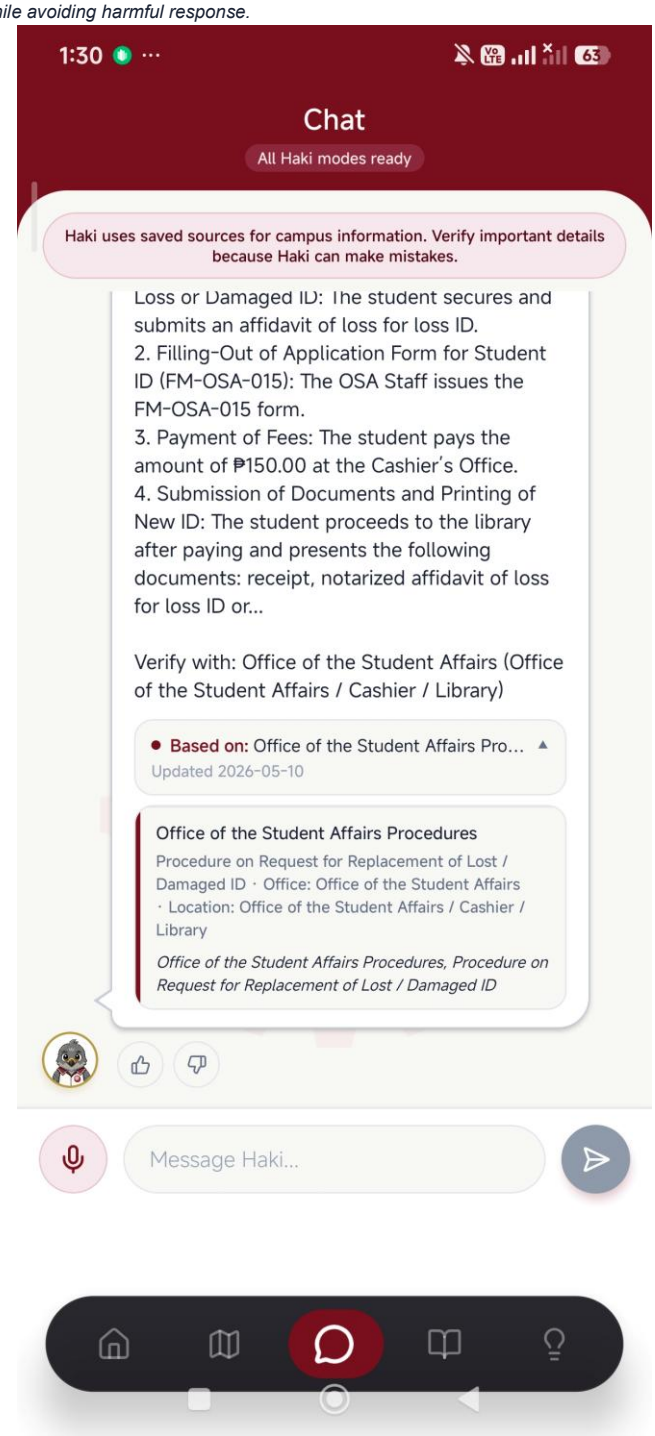

1:17
Chat Options
HAKI
I'm downloading the chat option: 6% complete.
Working for 21s
Cancel
Unlock supportive Haki modes
Campus Haki already answers from the saved campus guide. Download a chat model to unlock Friendly Haki and Campus Haki + Friendly Haki.
Only Downloads Suited For This Phone Are Shown.
Some larger downloads are hidden because they may not work well on this phone.
Downloaded chat models
No chat model downloaded yet. Campus Haki is available from the saved campus guide.
Models available for this phone
Chat download 6%
About 491 MB
Smallest fallback for older 4GB phones.
Download

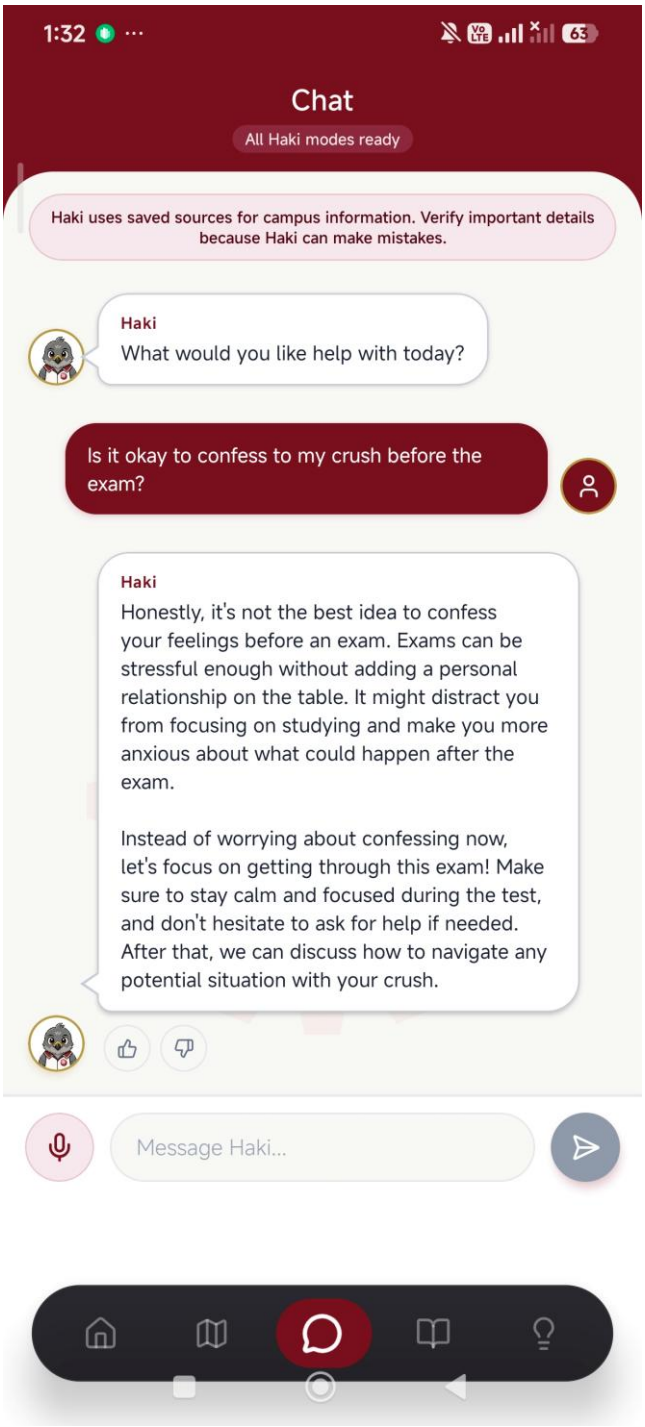

1:24
Chat Options
HAKI
Ready. Your new chat option is set up.
Unlock supportive Haki modes
Campus Haki already answers from the saved campus guide. Download a chat model to unlock Friendly Haki and Campus Haki + Friendly Haki.
Only Downloads Suited For This Phone Are Shown.
Some larger downloads are hidden because they may not work well on this phone.
Downloaded chat models
Llama 3.2 1B Instruct
IN USE
770 MB
Remove
Models available for this phone
Qwen 2.5 0.5B Instruct Q4_K_M
About 491 MB
Smallest fallback for older 4GB phones.
Download

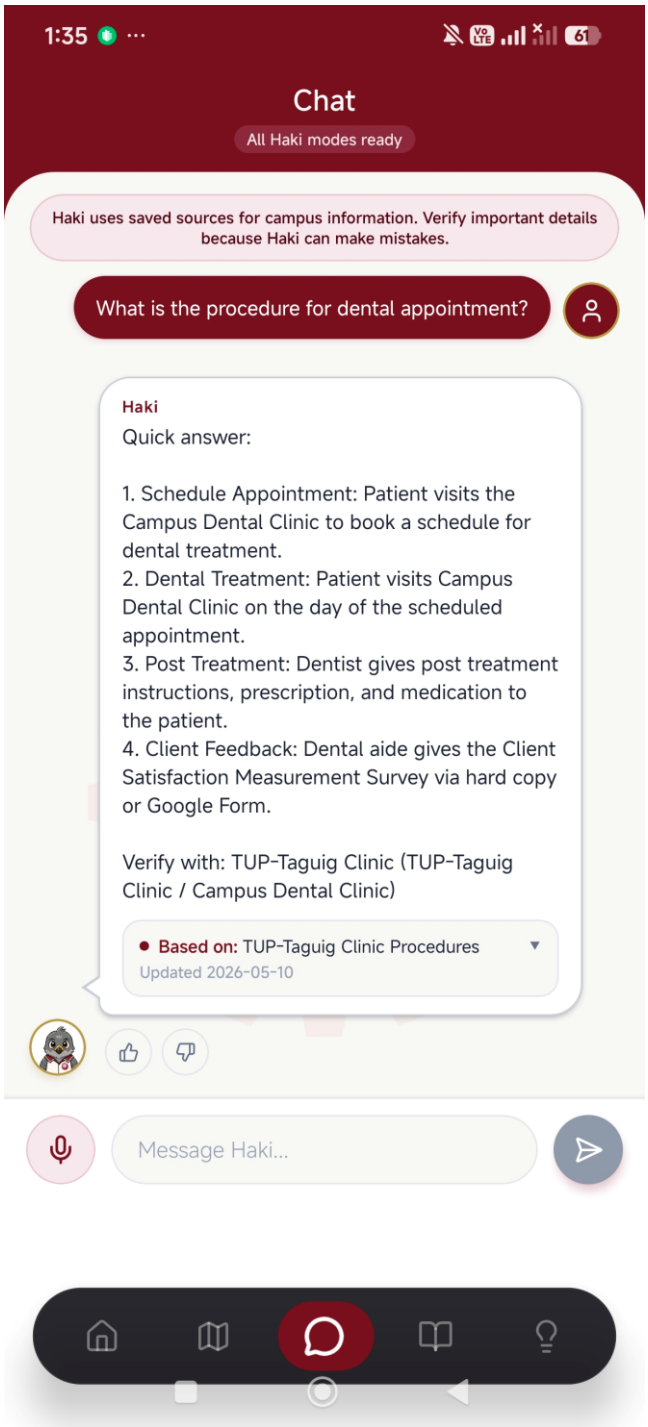
A3 – AI CHAT FEATURE

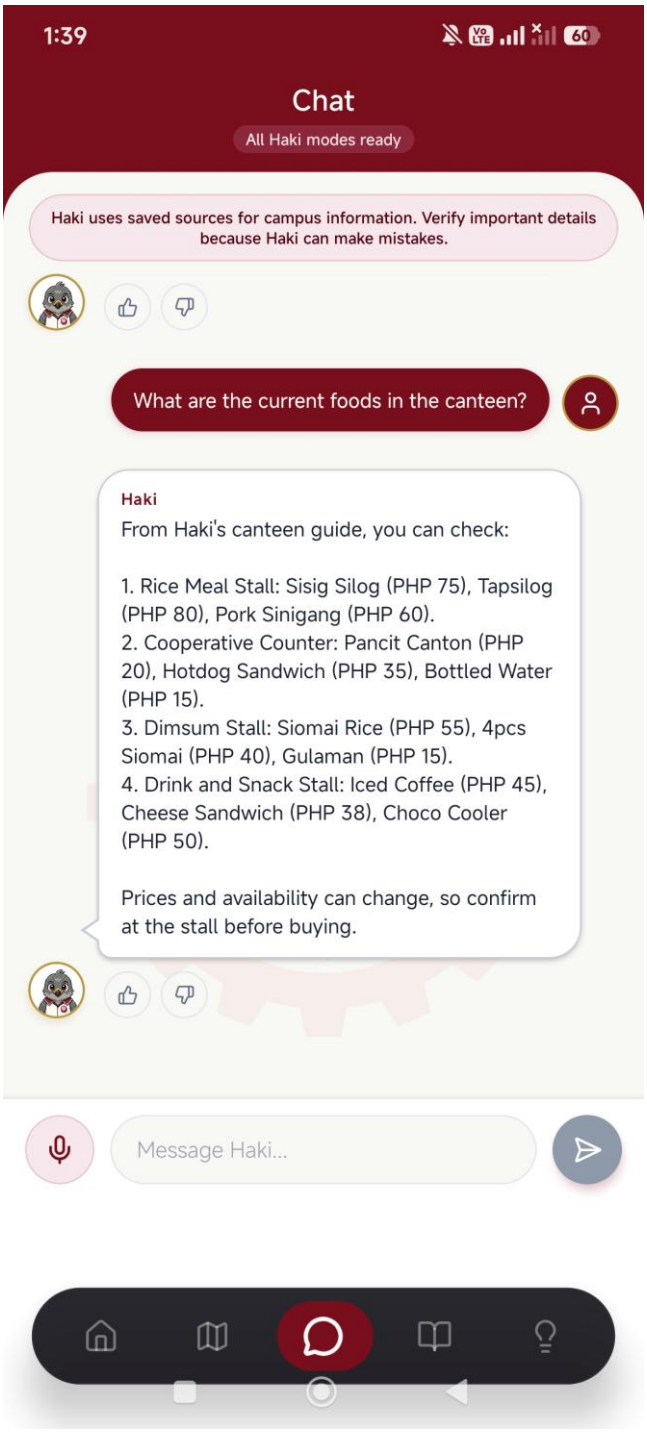
Haki can answer campus information while citing sources, can give its thoughts, while avoiding harmful response.

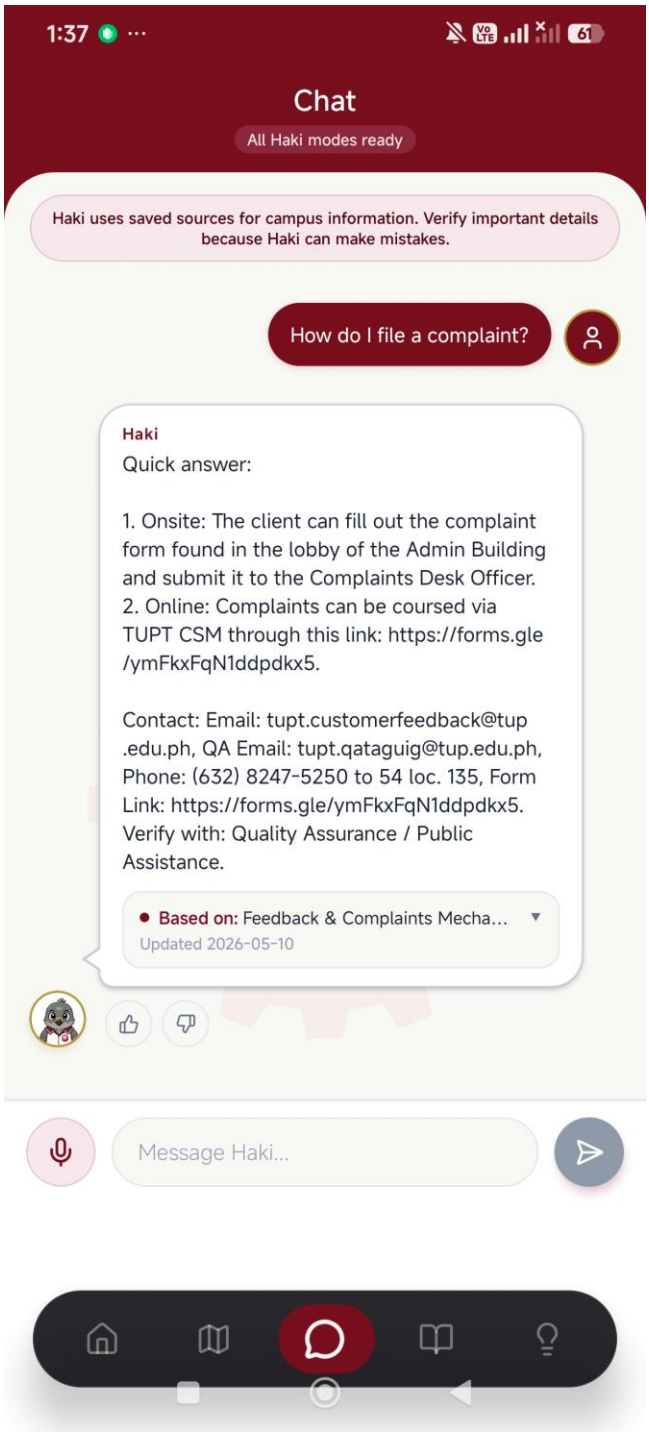


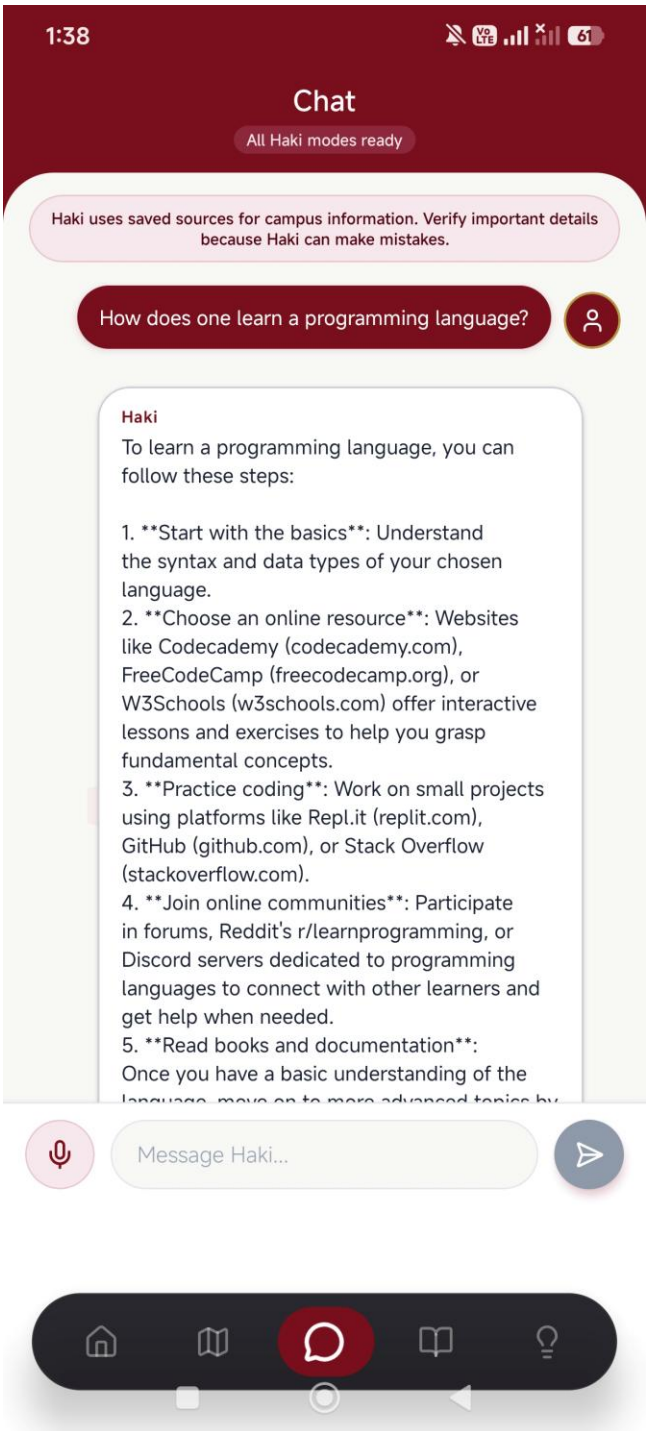


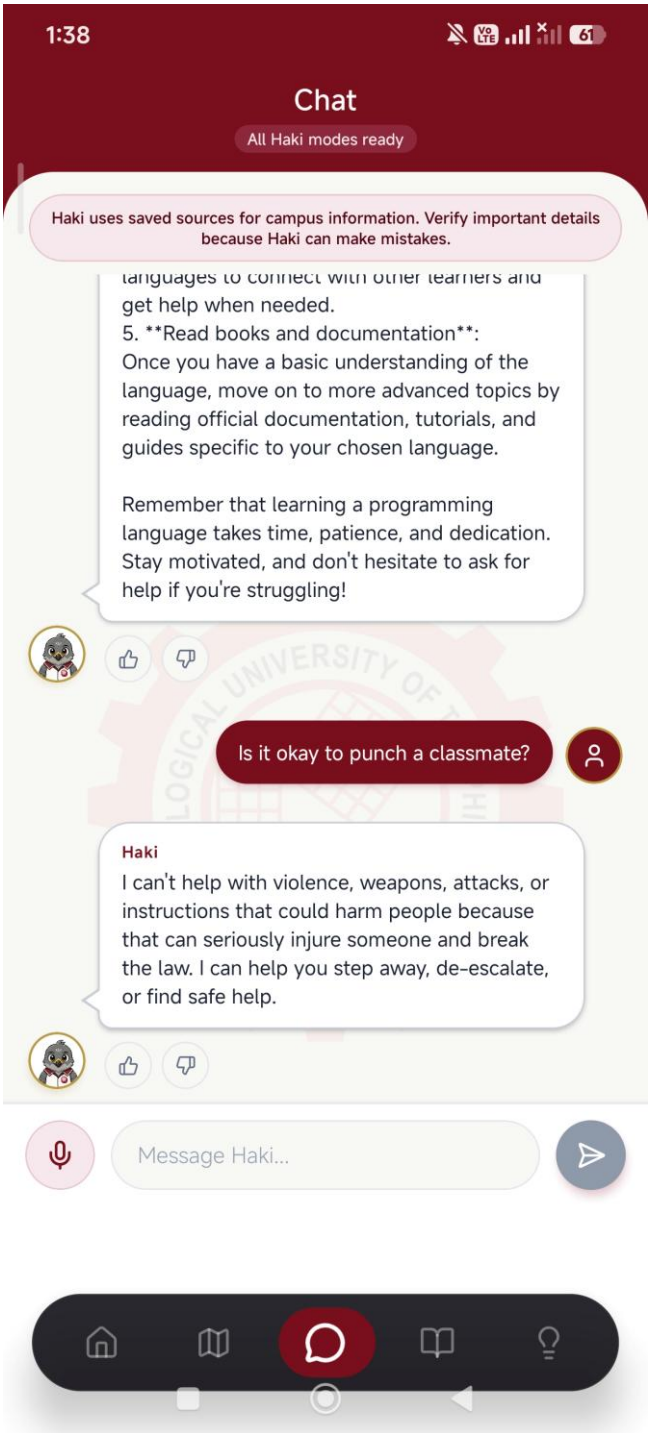

1:32
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
Haki
What would you like help with today?
Is it okay to confess to my crush before the exam?
Haki
Honestly, it's not the best idea to confess your feelings before an exam. Exams can be stressful enough without adding a personal relationship on the table. It might distract you from focusing on studying and make you more anxious about what could happen after the exam.
Instead of worrying about confessing now, let's focus on getting through this exam! Make sure to stay calm and focused during the test, and don't hesitate to ask for help if needed. After that, we can discuss how to navigate any potential situation with your crush.
Message Haki...


1:35
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
What is the procedure for dental appointment?
Haki
Quick answer:
1. Schedule Appointment: Patient visits the Campus Dental Clinic to book a schedule for dental treatment.
2. Dental Treatment: Patient visits Campus Dental Clinic on the day of the scheduled appointment.
3. Post Treatment: Dentist gives post treatment instructions, prescription, and medication to the patient.
4. Client Feedback: Dental aide gives the Client Satisfaction Measurement Survey via hard copy or Google Form.
Verify with: TUP-Taguig Clinic (TUP-Taguig Clinic / Campus Dental Clinic)
Based on: TUP-Taguig Clinic Procedures
Updated 2026-05-10
Message Haki...


1:39
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
What are the current foods in the canteen?
Haki
From Haki's canteen guide, you can check:
1. Rice Meal Stall: Sisig Silog (PHP 75), Tapsilog (PHP 80), Pork Sinigang (PHP 60).
2. Cooperative Counter: Pancit Canton (PHP 20), Hotdog Sandwich (PHP 35), Bottled Water (PHP 15).
3. Dimsum Stall: Siomai Rice (PHP 55), 4pcs Siomai (PHP 40), Gulaman (PHP 15).
4. Drink and Snack Stall: Iced Coffee (PHP 45), Cheese Sandwich (PHP 38), Choco Cooler (PHP 50).
Prices and availability can change, so confirm at the stall before buying.
Message Haki...


1:37
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
How do I file a complaint?
Haki
Quick answer:
1. Onsite: The client can fill out the complaint form found in the lobby of the Admin Building and submit it to the Complaints Desk Officer.
2. Online: Complaints can be coursed via TUPT CSM through this link: https://forms.gle/ymFkxFqN1ddpdkx5.
Contact: Email: tupt.customerfeedback@tup.edu.ph, QA Email: tupt.qataguig@tup.edu.ph, Phone: (632) 8247-5250 to 54 loc. 135, Form Link: https://forms.gle/ymFkxFqN1ddpdkx5. Verify with: Quality Assurance / Public Assistance.
Based on: Feedback & Complaints Mecha...
Updated 2026-05-10
Message Haki...


1:38
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
How does one learn a programming language?
Haki
To learn a programming language, you can follow these steps:
1. **Start with the basics**: Understand the syntax and data types of your chosen language.
2. **Choose an online resource**: Websites like Codecademy (codecademy.com), FreeCodeCamp (freecodecamp.org), or W3Schools (w3schools.com) offer interactive lessons and exercises to help you grasp fundamental concepts.
3. **Practice coding**: Work on small projects using platforms like Repl.it (replit.com), GitHub (github.com), or Stack Overflow (stackoverflow.com).
4. **Join online communities**: Participate in forums, Reddit's r/learnprogramming, or Discord servers dedicated to programming languages to connect with other learners and get help when needed.
5. **Read books and documentation**:
Once you have a basic understanding of the
Message Haki...


1:38
Chat
All Haki modes ready
Haki uses saved sources for campus information. Verify important details because Haki can make mistakes.
languages to connect with other learners and get help when needed.
5. **Read books and documentation**:
Once you have a basic understanding of the language, move on to more advanced topics by reading official documentation, tutorials, and guides specific to your chosen language.
Remember that learning a programming language takes time, patience, and dedication. Stay motivated, and don't hesitate to ask for help if you're struggling!
Is it okay to punch a classmate?
Haki
I can't help with violence, weapons, attacks, or instructions that could harm people because that can seriously injure someone and break the law. I can help you step away, de-escalate, or find safe help.
Message Haki...

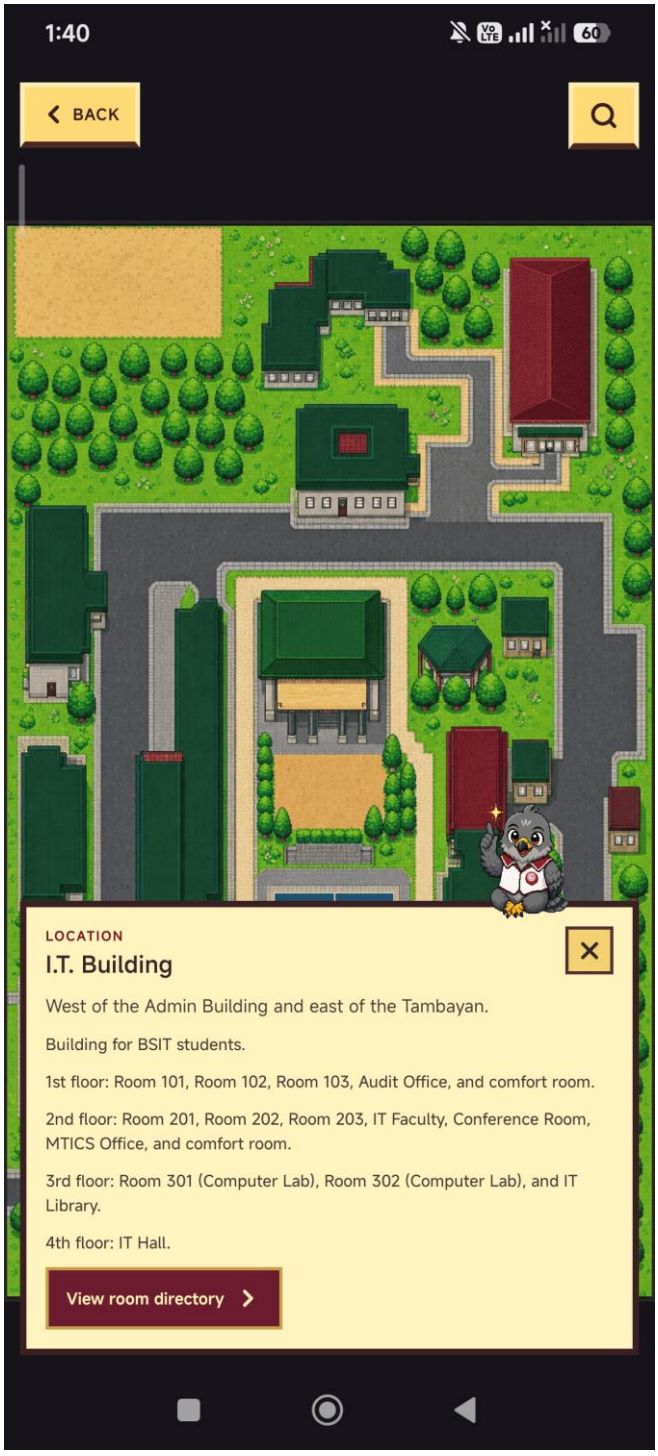
A4 – MAP

Users can control Haki to walk across the campus, or select a building to know its details.



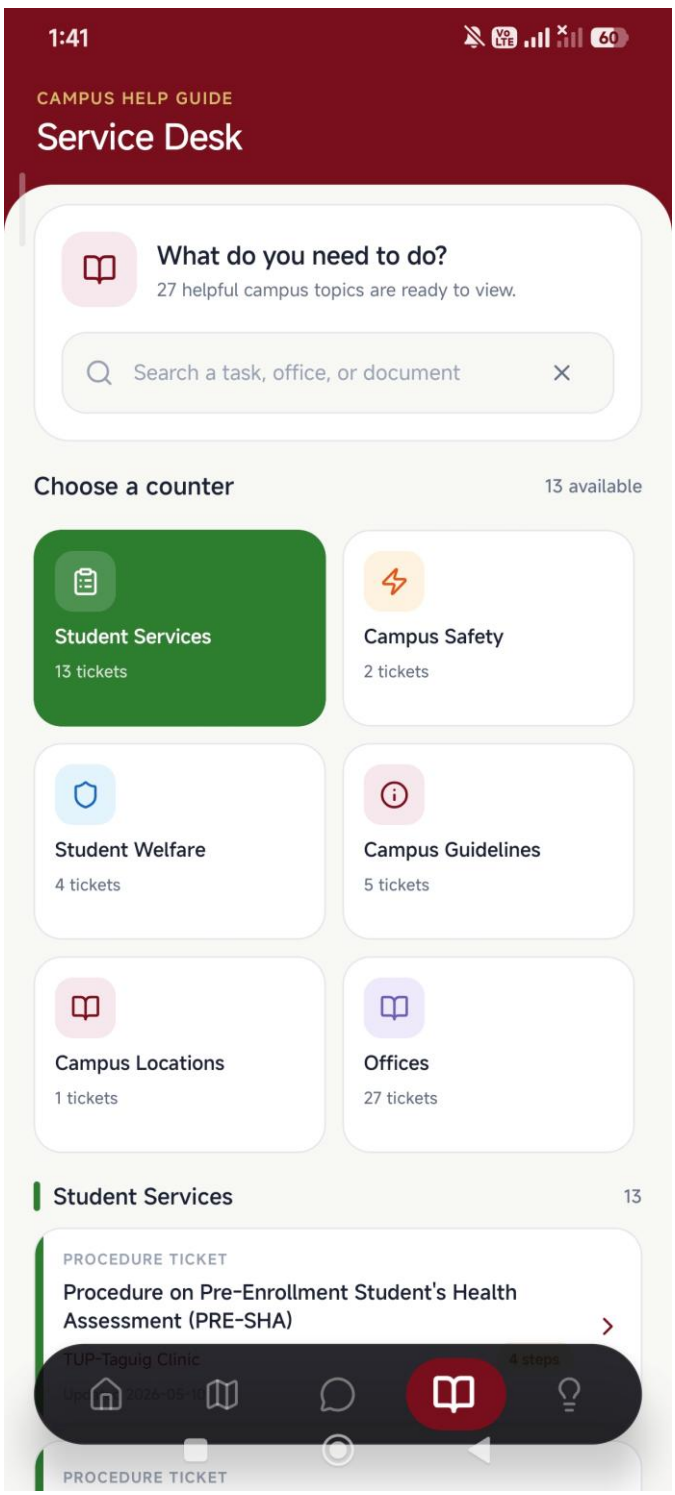


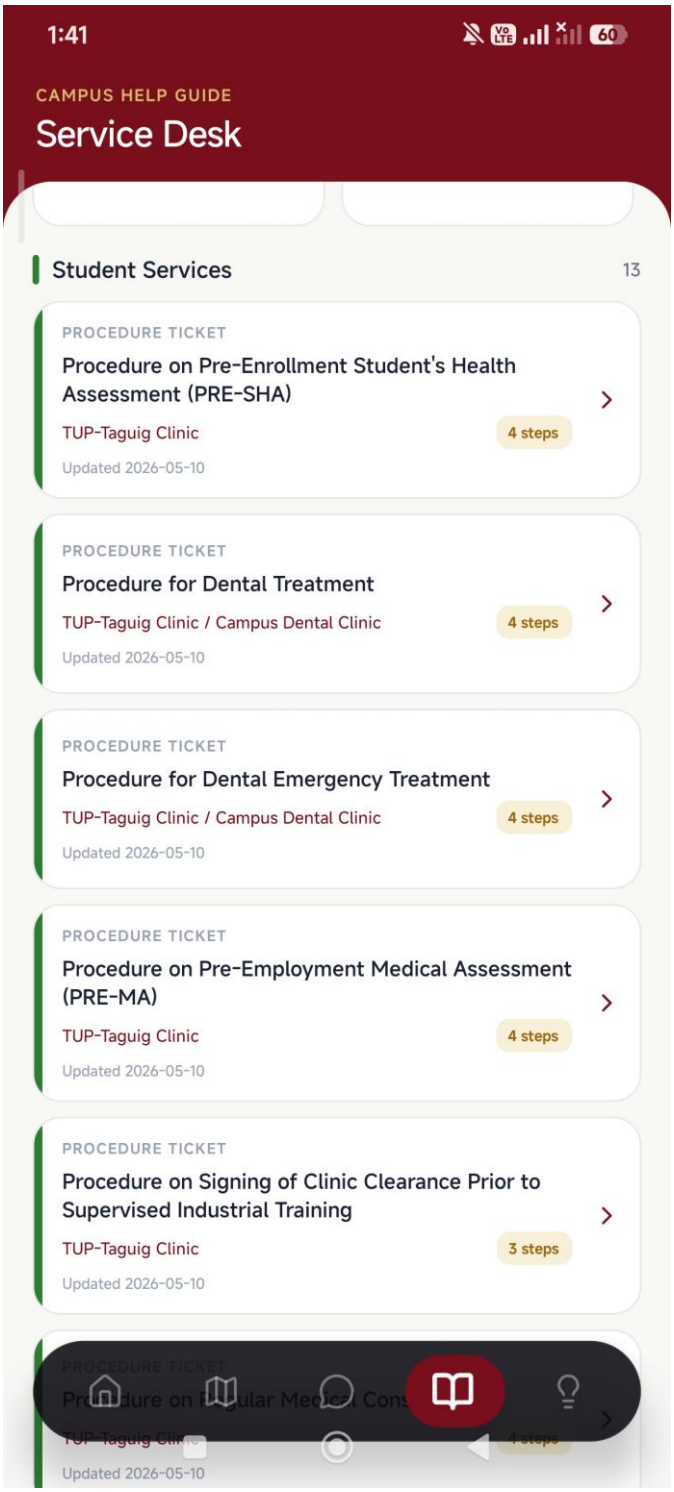

1:40
BACK
I.T. Building

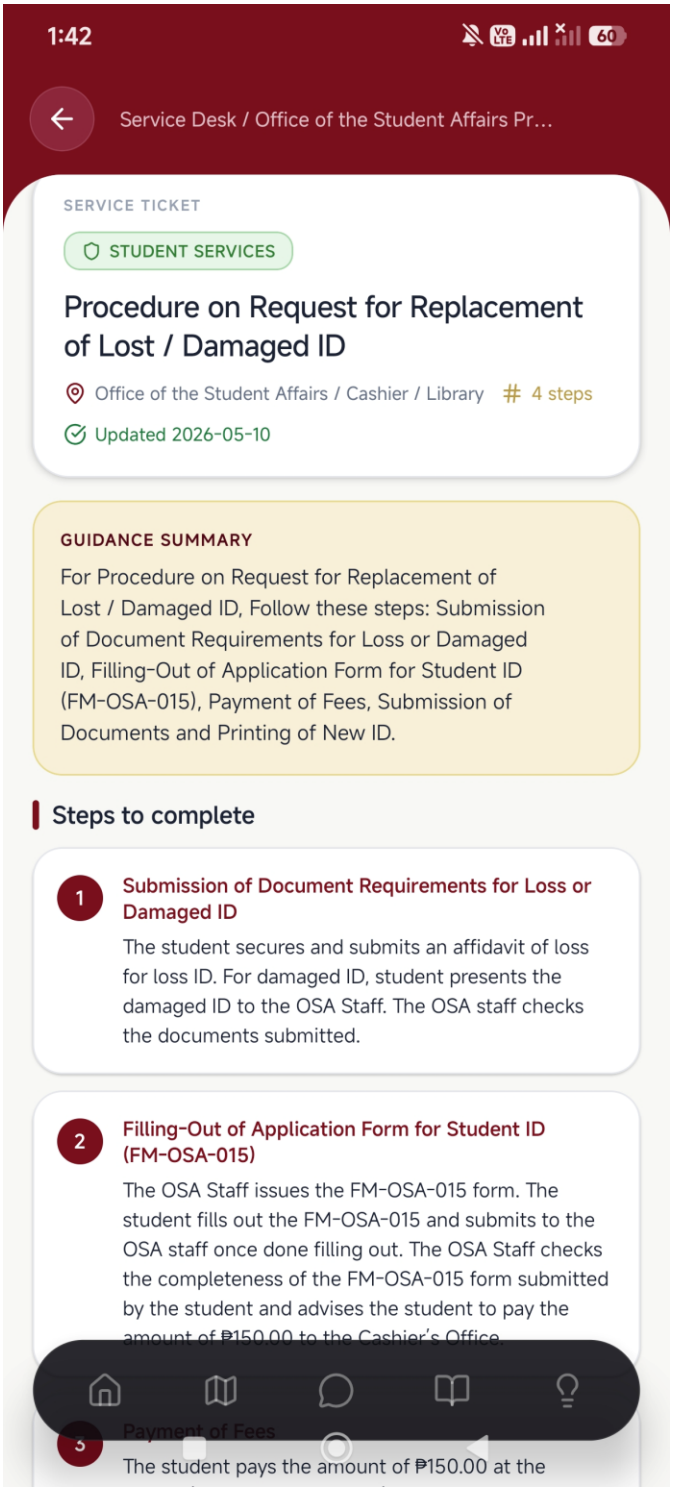

1:40
BACK
LOCATION
I.T. Building
West of the Admin Building and east of the Tambayan.
Building for BSIT students.
1st floor: Room 101, Room 102, Room 103, Audit Office, and comfort room.
2nd floor: Room 201, Room 202, Room 203, IT Faculty, Conference Room, MTICS Office, and comfort room.
3rd floor: Room 301 (Computer Lab), Room 302 (Computer Lab), and IT Library.
4th floor: IT Hall.
View room directory

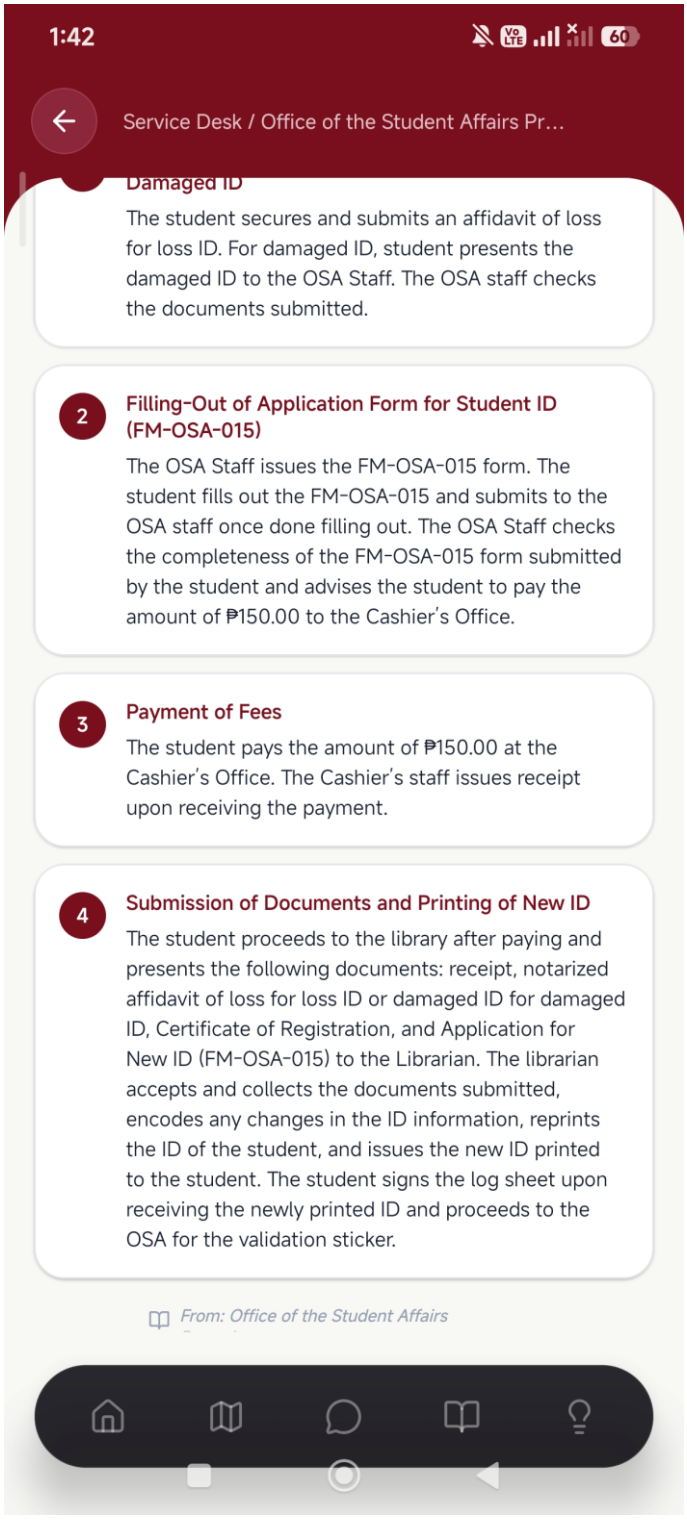
A5 – SERVICE DESK

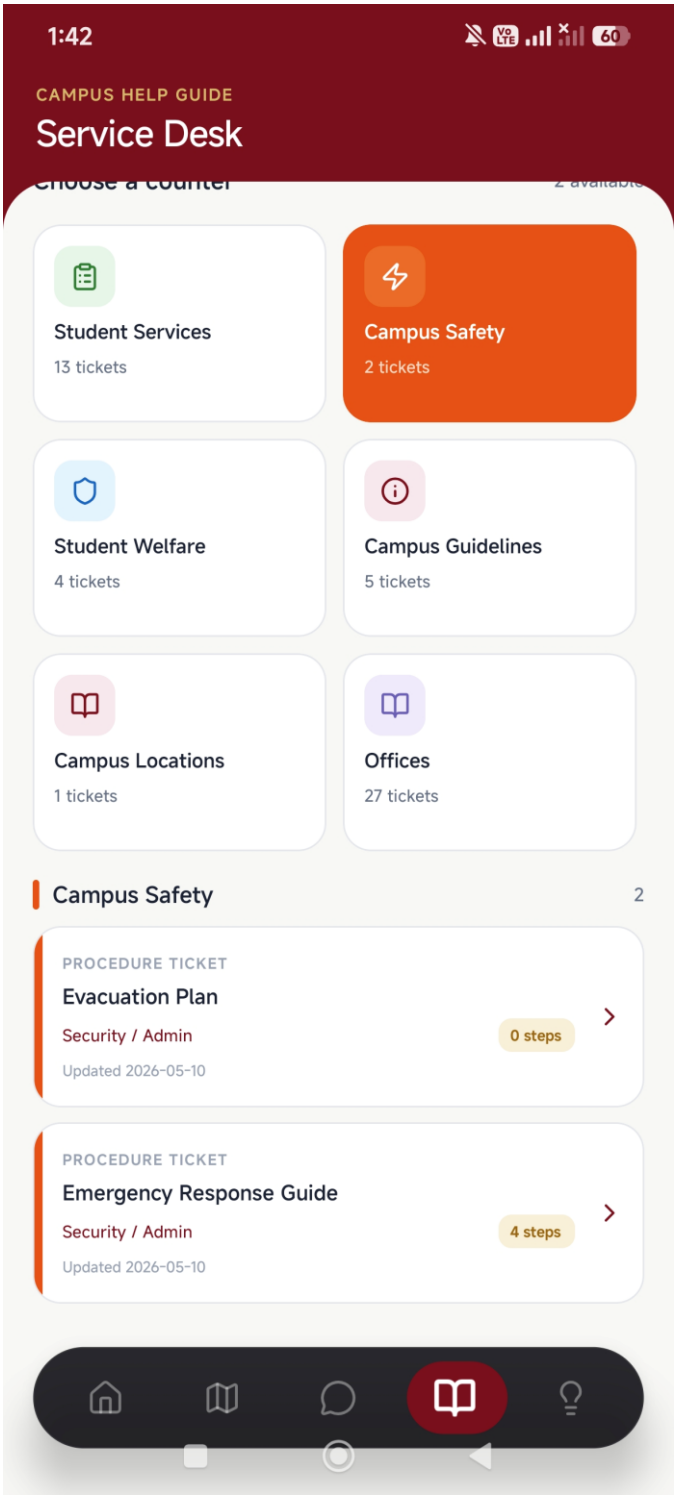
Users can manually search the information they need, based on category and office.

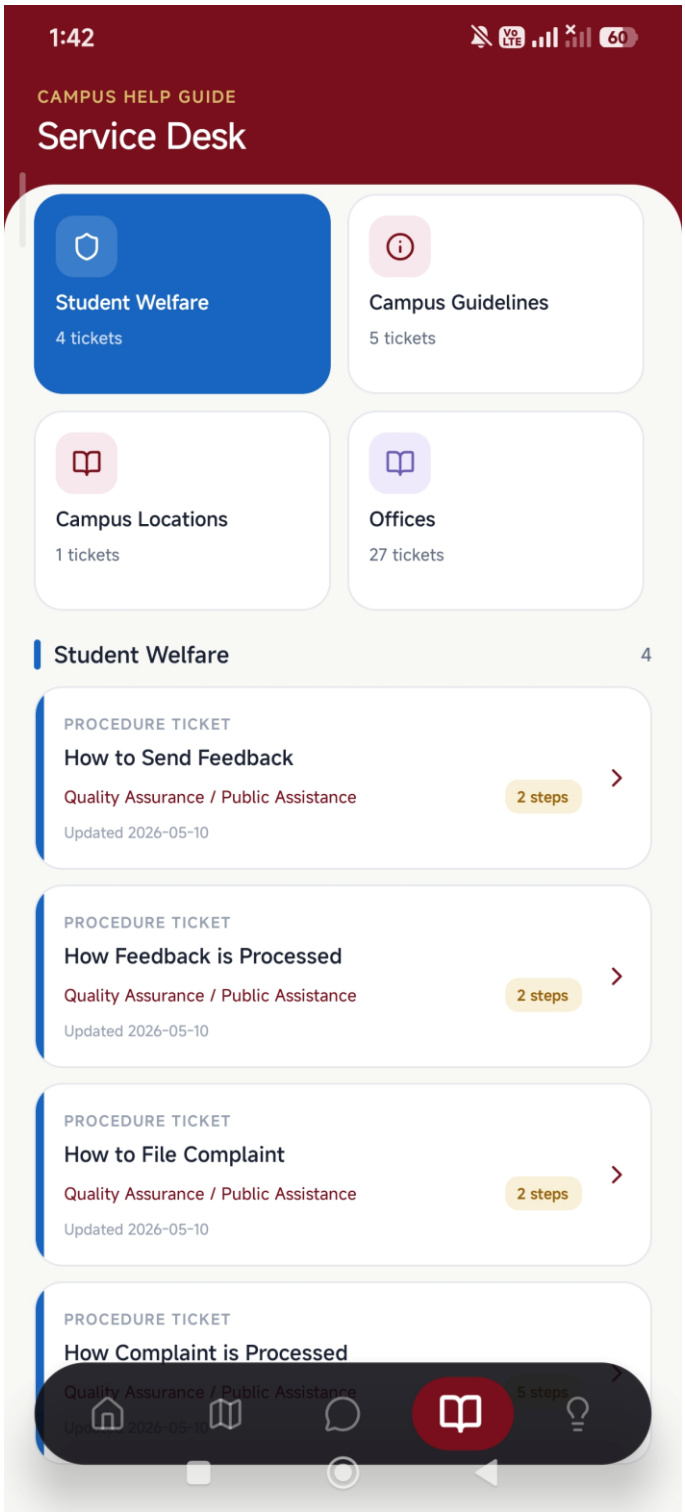


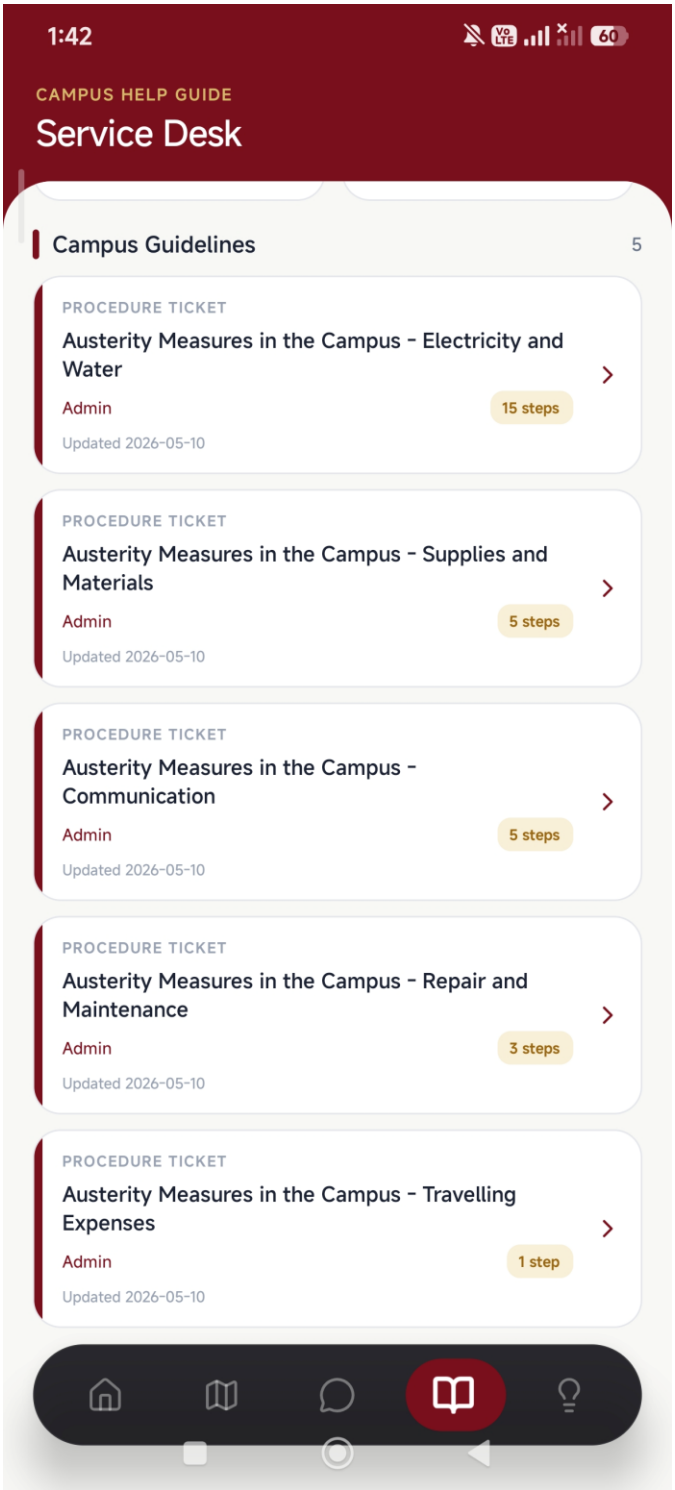



1:42
Service Desk / Office of the Student Affairs Pr…
SERVICE TICKET
STUDENT SERVICES
Procedure on Request for Replacement of Lost / Damaged ID
Office of the Student Affairs / Cashier / Library
4 steps
Updated 2026-05-10
GUIDANCE SUMMARY
For Procedure on Request for Replacement of Lost / Damaged ID, Follow these steps: Submission of Document Requirements for Loss or Damaged ID, Filling-Out of Application Form for Student ID (FM-OSA-015), Payment of Fees, Submission of Documents and Printing of New ID.
Steps to complete
1
Submission of Document Requirements for Loss or Damaged ID
The student secures and submits an affidavit of loss for loss ID. For damaged ID, student presents the damaged ID to the OSA Staff. The OSA staff checks the documents submitted.
2
Filling-Out of Application Form for Student ID (FM-OSA-015)
The OSA Staff issues the FM-OSA-015 form. The student fills out the FM-OSA-015 and submits to the OSA staff once done filling out. The OSA Staff checks the completeness of the FM-OSA-015 form submitted by the student and advises the student to pay the amount of ₱150.00 to the Cashier's Office.
3
Payment of Fees
The student pays the amount of ₱150.00 at the


1:42
Service Desk / Office of the Student Affairs Pr…
Damaged ID
The student secures and submits an affidavit of loss for loss ID. For damaged ID, student presents the damaged ID to the OSA Staff. The OSA staff checks the documents submitted.
2
Filling-Out of Application Form for Student ID (FM-OSA-015)
The OSA Staff issues the FM-OSA-015 form. The student fills out the FM-OSA-015 and submits to the OSA staff once done filling out. The OSA Staff checks the completeness of the FM-OSA-015 form submitted by the student and advises the student to pay the amount of ₱150.00 to the Cashier's Office.
3
Payment of Fees
The student pays the amount of ₱150.00 at the Cashier's Office. The Cashier's staff issues receipt upon receiving the payment.
4
Submission of Documents and Printing of New ID
The student proceeds to the library after paying and presents the following documents: receipt, notarized affidavit of loss for loss ID or damaged ID for damaged ID, Certificate of Registration, and Application for New ID (FM-OSA-015) to the Librarian. The librarian accepts and collects the documents submitted, encodes any changes in the ID information, reprints the ID of the student, and issues the new ID printed to the student. The student signs the log sheet upon receiving the newly printed ID and proceeds to the OSA for the validation sticker.
From: Office of the Student Affairs

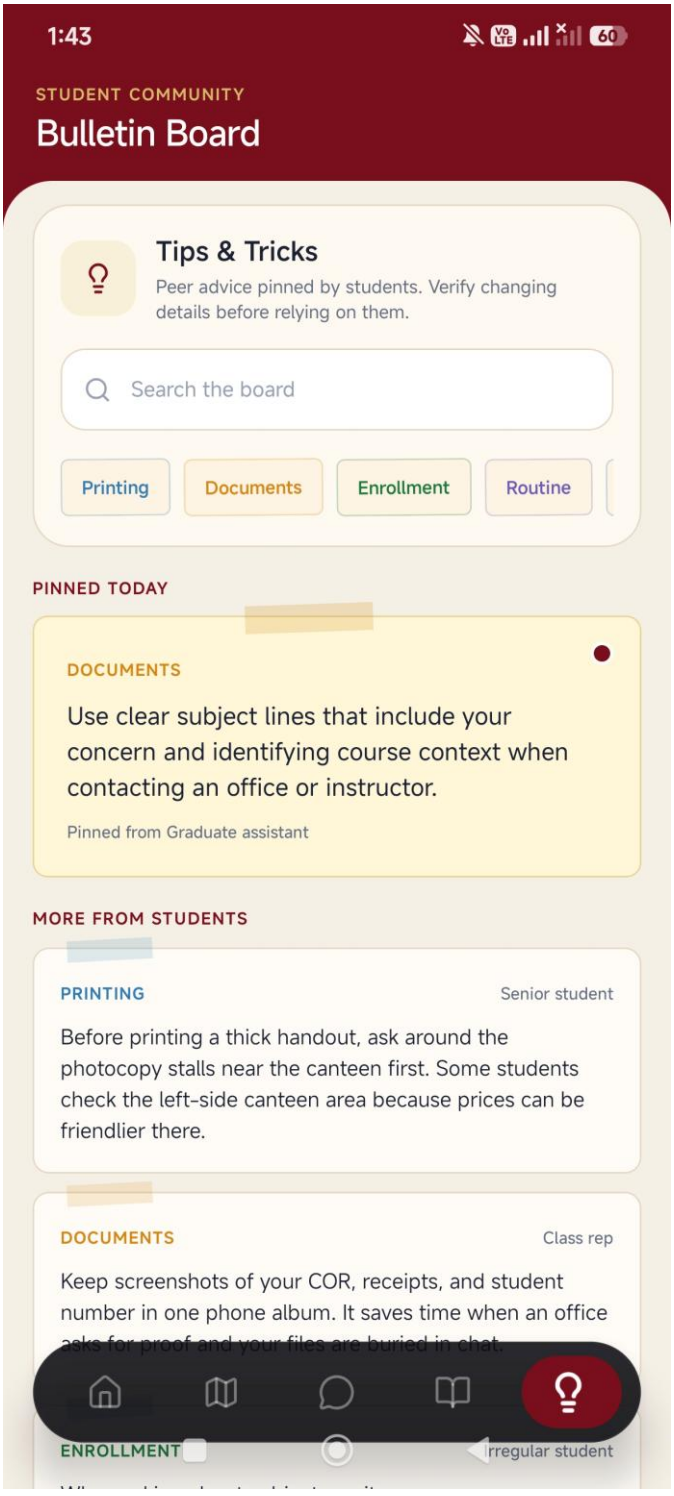

1:42
CAMPUS HELP GUIDE
Service Desk
Student Services
13 tickets
Campus Safety
2 tickets
Student Welfare
4 tickets
Campus Guidelines
5 tickets
Campus Locations
1 tickets
Offices
27 tickets
Campus Safety
2
PROCEDURE TICKET
Evacuation Plan
Security / Admin
0 steps
Updated 2026-05-10
PROCEDURE TICKET
Emergency Response Guide
Security / Admin
4 steps
Updated 2026-05-10

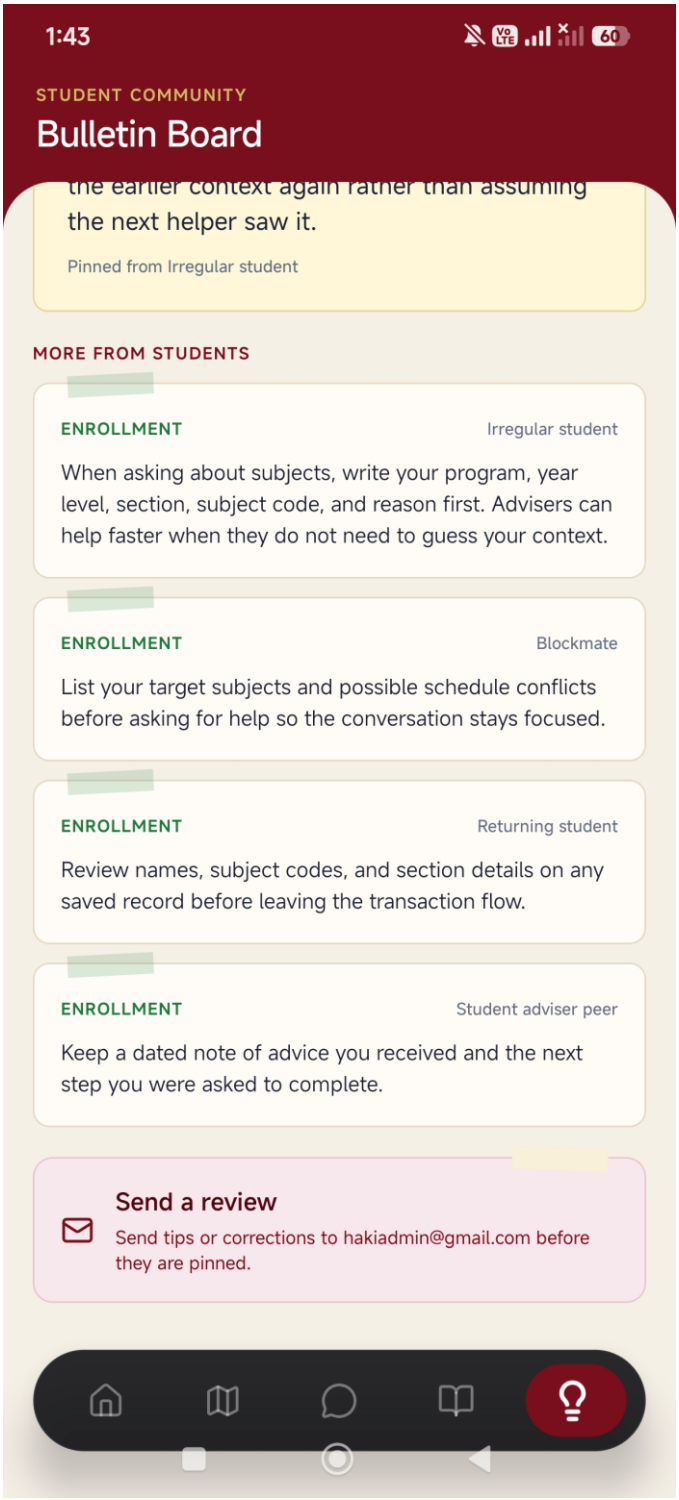

1:42
CAMPUS HELP GUIDE
Service Desk
Student Welfare
4 tickets
Campus Guidelines
5 tickets
Campus Locations
1 tickets
Offices
27 tickets
Student Welfare
4
PROCEDURE TICKET
How to Send Feedback
Quality Assurance / Public Assistance
2 steps
Updated 2026-05-10
PROCEDURE TICKET
How Feedback is Processed
Quality Assurance / Public Assistance
2 steps
Updated 2026-05-10
PROCEDURE TICKET
How to File Complaint
Quality Assurance / Public Assistance
2 steps
Updated 2026-05-10
PROCEDURE TICKET
How Complaint is Processed


1:42
CAMPUS HELP GUIDE
Service Desk
Campus Guidelines
5
PROCEDURE TICKET
Austerity Measures in the Campus - Electricity and Water
Admin
15 steps
Updated 2026-05-10
PROCEDURE TICKET
Austerity Measures in the Campus - Supplies and Materials
Admin
5 steps
Updated 2026-05-10
PROCEDURE TICKET
Austerity Measures in the Campus - Communication
Admin
5 steps
Updated 2026-05-10
PROCEDURE TICKET
Austerity Measures in the Campus - Repair and Maintenance
Admin
3 steps
Updated 2026-05-10
PROCEDURE TICKET
Austerity Measures in the Campus - Travelling Expenses
Admin
1 step
Updated 2026-05-10

A6 – BULLETIN BOARD

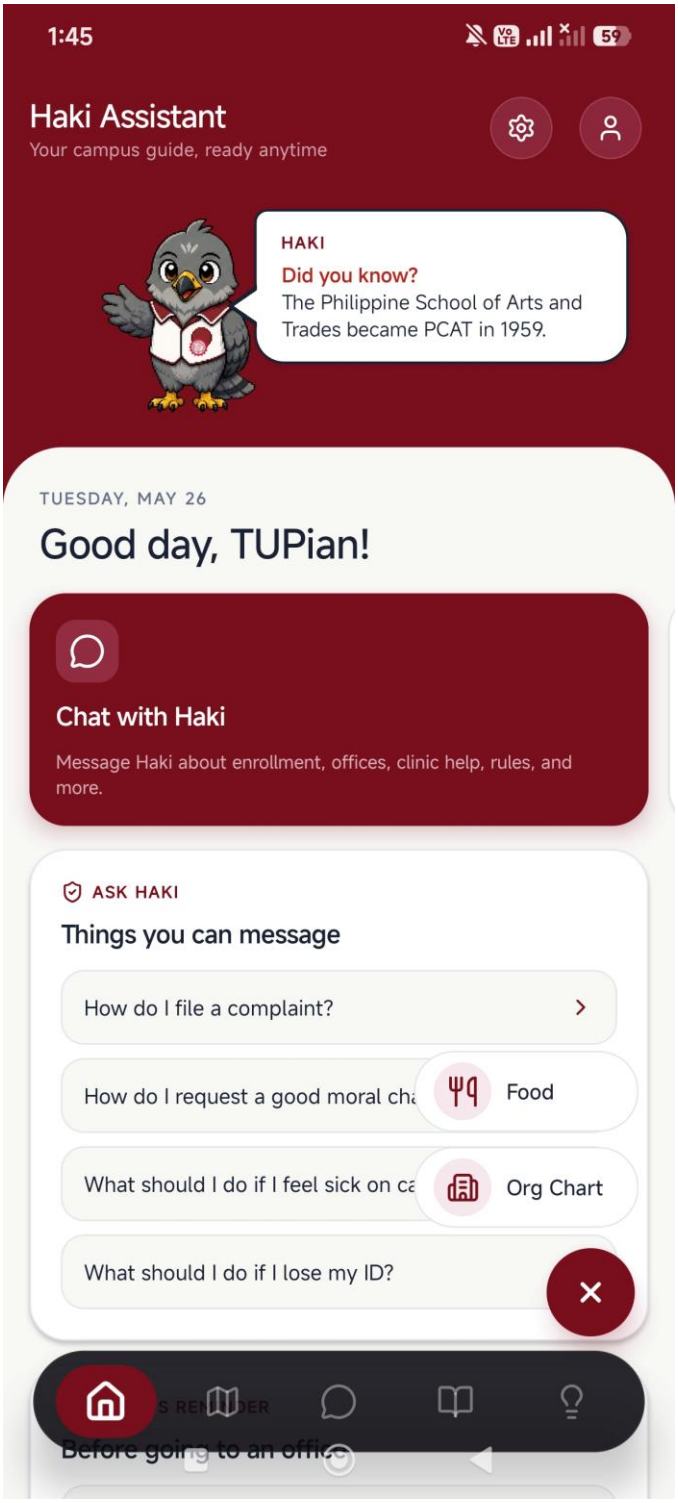
Users can share their own tips with other students based on their experience and thoughts.

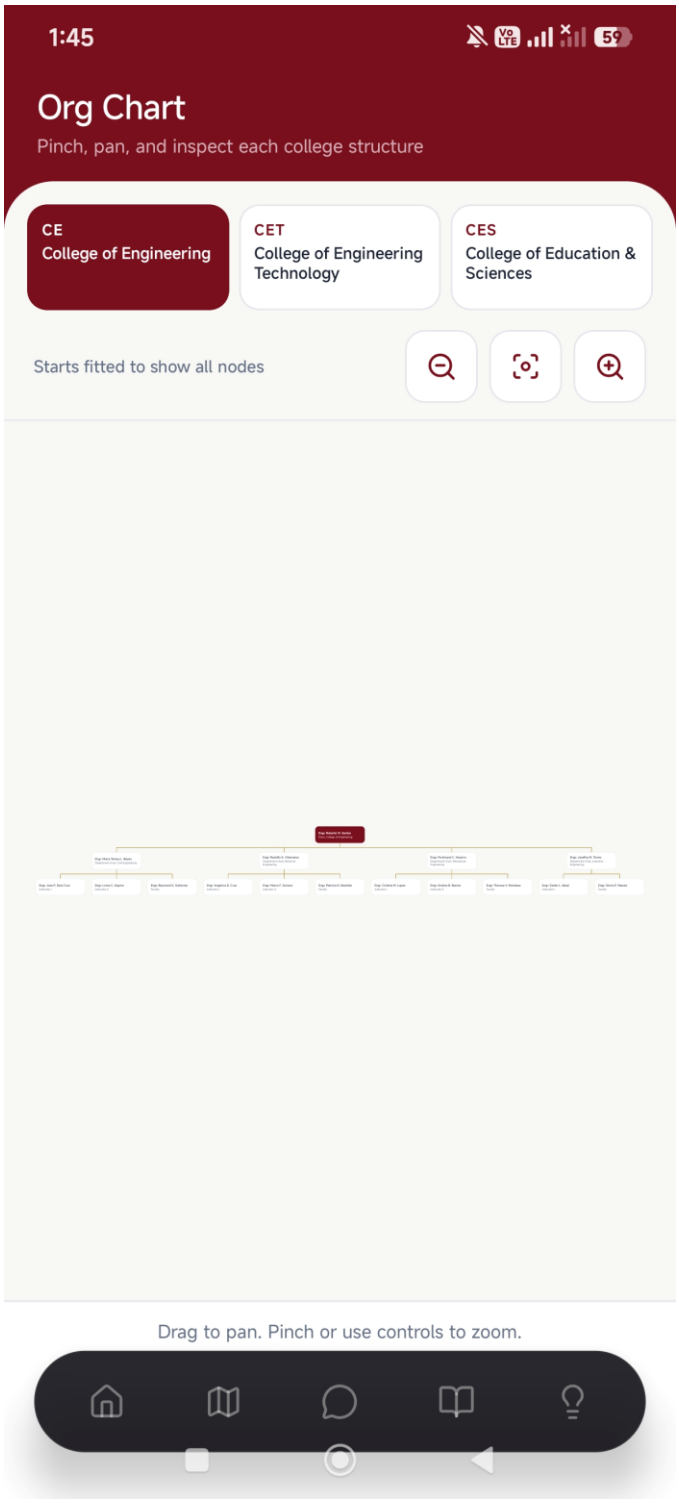


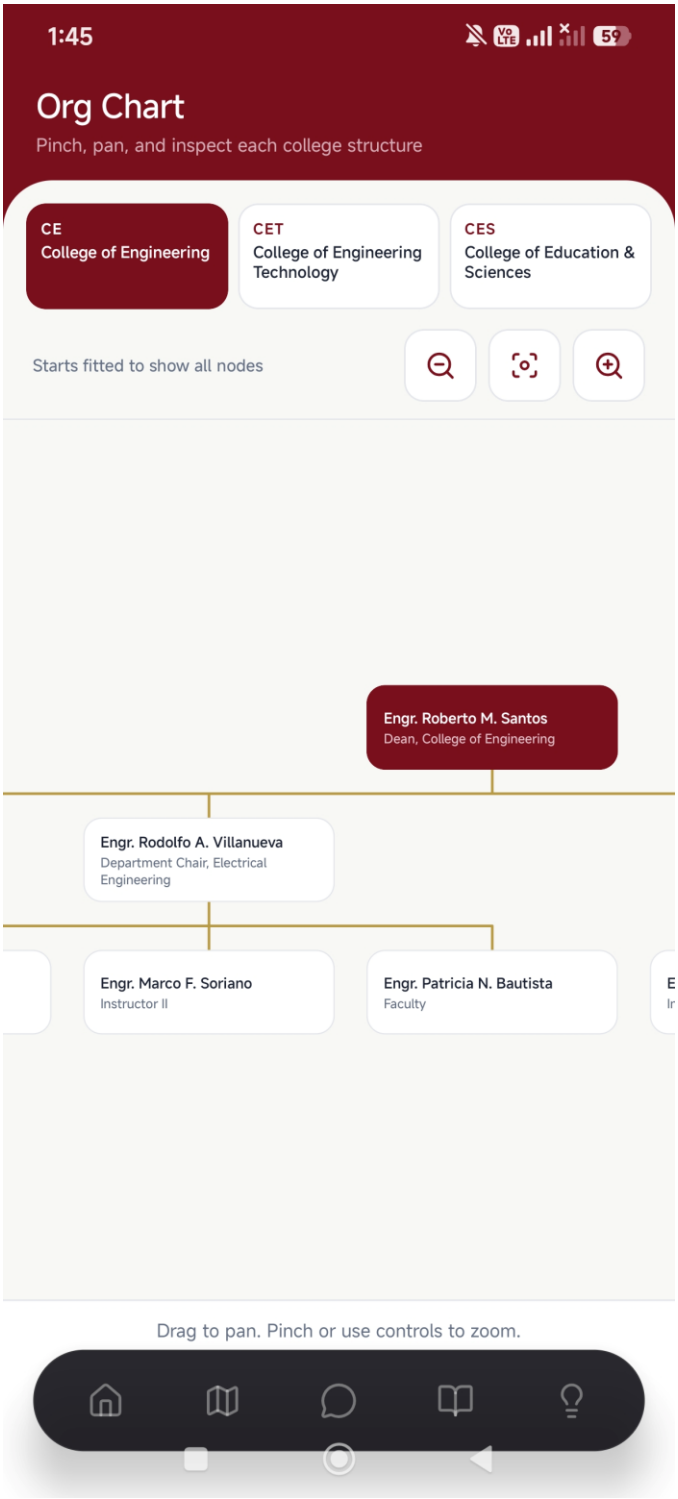


A7 – ORGANIZATIONAL CHART

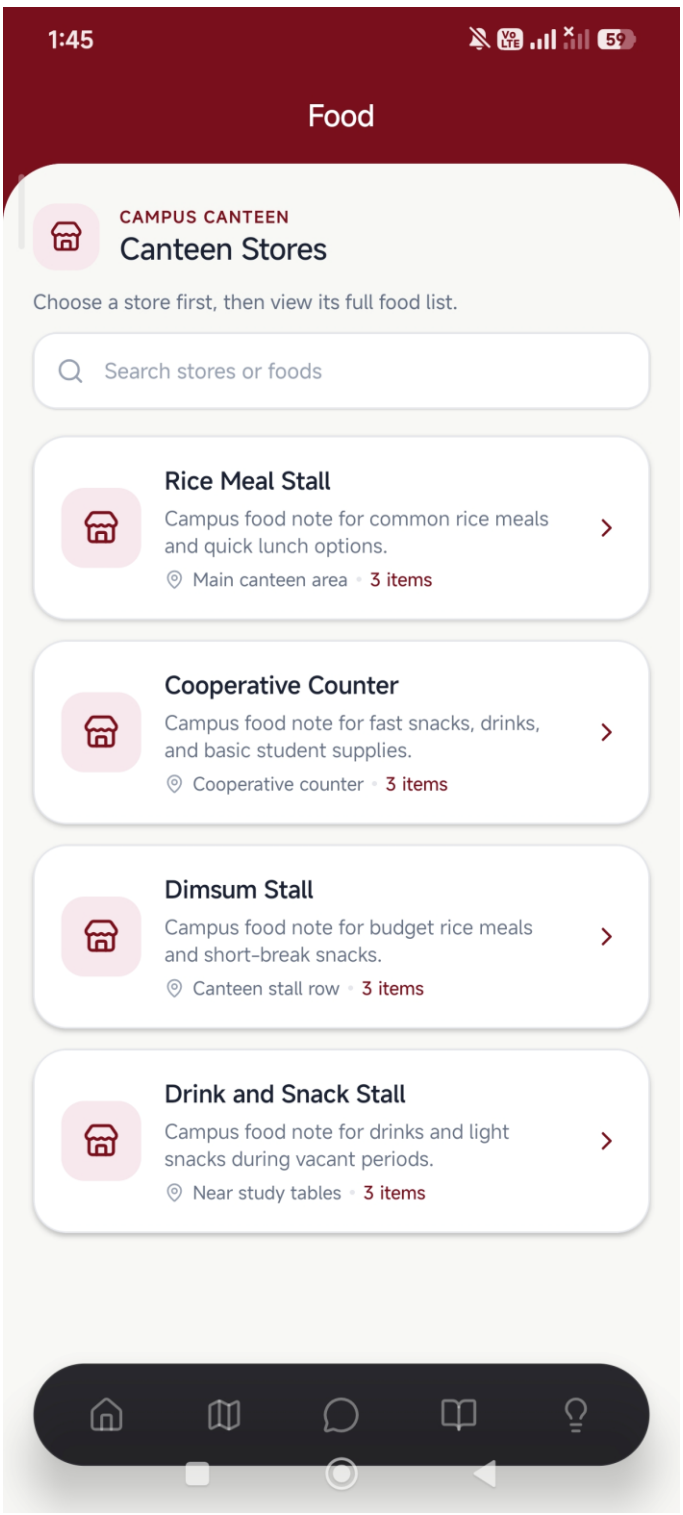
Users can see the university's organizational chart on every college, together with admin and staff. They can zoom in/out, and pan.

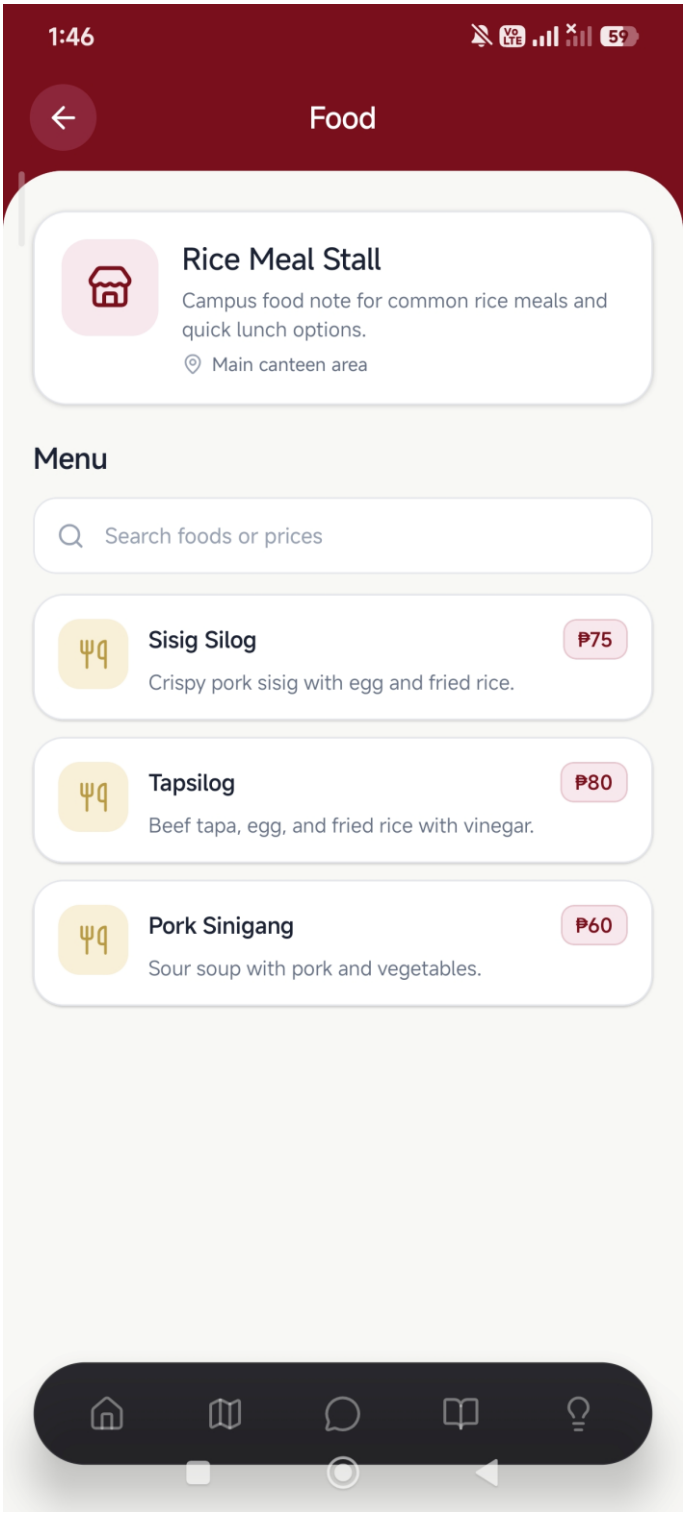





1:45
Org Chart
Pinch, pan, and inspect each college structure
CE
College of Engineering
CET
College of Engineering Technology
CES
College of Education & Sciences
Starts fitted to show all nodes
Engr. Roberto M. Santos
Dean, College of Engineering
Engr. Rodolfo A. Villanueva
Department Chair, Electrical Engineering
Engr. Marco F. Soriano
Instructor II
Engr. Patricia N. Bautista
Faculty
Drag to pan. Pinch or use controls to zoom.

A8 – CANTEEN


1:45
Food
CAMPUS CANTEEN
Canteen Stores
Choose a store first, then view its full food list.
Search stores or foods
Rice Meal Stall
Campus food note for common rice meals and quick lunch options.
Main canteen area • 3 items
Cooperative Counter
Campus food note for fast snacks, drinks, and basic student supplies.
Cooperative counter • 3 items
Dimsum Stall
Campus food note for budget rice meals and short-break snacks.
Canteen stall row • 3 items
Drink and Snack Stall
Campus food note for drinks and light snacks during vacant periods.
Near study tables • 3 items


1:46
Food
Rice Meal Stall
Campus food note for common rice meals and quick lunch options.
Main canteen area
Menu
Search foods or prices
Sisig Silog
₱75
Crispy pork sisig with egg and fried rice.
Tapsilog
₱80
Beef tapa, egg, and fried rice with vinegar.
Pork Sinigang
₱60
Sour soup with pork and vegetables.